Development of a Software Package Designed to Support Distance Education for Disabled People

Vyacheslav I. Voronov, Ksenia V. Genchel, Lilia I. Voronova, Maria D. Travina
Department of Intelligent Control Systems and Automation
Moscow Technical University of Communications and Informatics (MTUCI)
Moscow, Russia
isua.kaf@ya.ru

***Abstract*—Today, the society pays much attention to the problem of adapting people with disabilities, in particular, to improving the quality of their education and ways of teaching. The article describes the development of the software package "Surdotelephone" with the help of intelligent data processing technologies, such as machine learning methods, to help people with hearing impairments who receive distance education. The project provides recognition by dactyl of sign language.**

Keywords— recognition; sign language, communication, smart education, kNN, SVM, logistic regression

I. INTRODUCTION

Currently, special attention is paid to improving the quality of life of people with disabilities, in particular the expansion of methods and techniques of education. In the learning process, active communication between the teacher and the student is necessary, which is difficult when the student is a hard-of-hearing person.

Communication is a two-way process, which is the transfer of information from the "speaker" to the "deaf-mute" (type I) and the translation of the sign language into symbolic text or speech (type II). We can designate the first type as a direct problem, and the second as a reverse one.

To the first type translation, you can now use the already widespread computer technology – п. speech recognition. That technology can convert voice speech in visual symbols. The formalization of the second type communication and the implementation of computer programs that provide recognition and "translation" of the sign language are a much more complex task requiring the use of intelligent methods of processing information. In this regard, the development of communication systems for social accessibility for people with disabilities on the basis of methods of machine learning is relevant.

The language spoken by people with hearing problems is called sign language. This is a very complex language that has many dialects. They talk in this language, usually by two hands, but many things can be explained with one hand.

The article deals with the inverse problem associated with the development of computer-based methods and their implementation in the software package, supporting the learning process of a hearing impaired students.

Fig. 1 shows the Russian dactylic alphabet. Part of it is represented by static gestures, and the other - by dynamic gestures. Thus, the gestures in Fig. 2 are divided into 23 static gestures and 9 dynamic gestures.



Fig. 1. The Russian Dactyl Alphabet [1]

Alphabet Letters Corresponding to Dactyls	**Category**
А, В, Г, Е, Ж, И, Л, М, Н, О, П, Р, С, Т, У, Ф, Х, Ч, Щ, Ы, Э, Ю, Я	Static
Б, Д, З, Й, К, Ц, Щ, Ь, Ъ	Dinamic

Fig. 2. Categories of the Russian Dactyl language

Functionality of the developing software package includes two tasks. The first task is translating the speaker's words into visual symbols (or gestures with the help of an avatar) and displaying them on screen of a hard of hearing student. The second task, as already noted, is a more complex and still not completely solved problem – the gesture recognition of and its machine translation into text / scoring. The solution of this problem is dealt with at the Department of Intelligent Systems in the management and automation of the Moscow Technical University of Communications and Informatics. The software complex was named "Surdotelephone" [2-4]. The idea of the program "Surdotelephone" consists in the following: a student (hard of hearing or deaf person) phone to the teacher through the appropriate program with a video translation, for example, on a smartphone and communicates with it using a set of gestures. The teacher's speech is translating into visual symbols (or gestures), and the project's task is to provide recognition of gestures of a hearing impaired student, transfer information and display it on the screen of a teacher.

978-1-5386-6757-6/18/$31.00 ©2018 IEEE

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 11:37:21 UTC from IEEE Xplore. Restrictions apply.

II. Scenarios Of The Levels Of The Model For Recognizing The Complex Structures Of Sign Language

There are a several types of complicated structures of sign language: dactyl (one letter of the dactyl alphabet), dactyl word (a word consisting of separate letters) and a full gesture that can denote both a word and a phrase. The following is the description of the levels scenarios of the model for recognizing the more complex sign language structures depicted using the BPwin computer simulation program.

In Fig.3. The model of recognition of static dactyls of the Russian sign language is presented:

- Video of the dactyl gesture is supposed to be divided into frames with a frequency of 1 frame per second.
- The frame will be translate into a format of 640 x 480 pixels.
- Then the image must be converted into a pixel intensity matrix and the matrix transformed into a vector.
- The vector is input to the gesture recognition program.
- If the gesture is recognized correctly, then the identified character will be displayed on the screen.
- In the case of recognition errors, you will be asked to write what this gesture means, and its image will be recorded in the database.

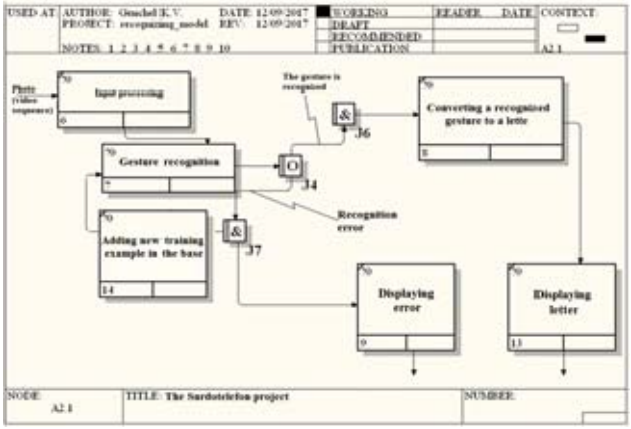


Fig. 3. Static Dactylic Recognition Model

Fig. 4 shows the model for recognizing dactyl words of sign language. From the above model of recognition of static dactyl, it differs in that, as long as one dactilem is recognized, the next dactyl gesture will be applied to the input, the processing will be parallel. This will continue until there is a gesture signifying the end of the word. In case of recognition error, a question mark will be displayed on the side of the unrecognized gesture. Will be invited to put a gesture in the database. If the word is correctly recognized, it will be displayed.

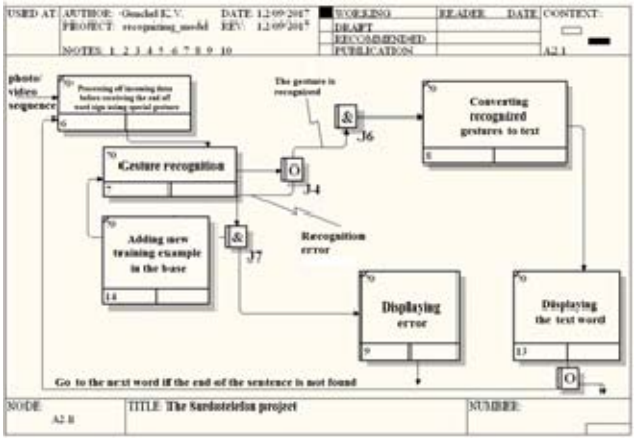


Fig. 4. Model of dactyl words of sign language recognition

Below is a model for recognizing dynamic gestures.

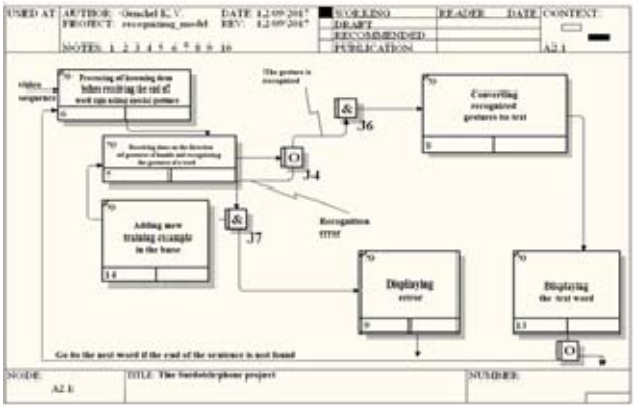


Fig. 5. Model of dynamic gestures recognition

Since this model involves recognizing the gesture in motion, after segmenting the video, the frames will be combined into packets, each of which will contain one gesture each. The dynamics of the gesture is tracked by these packets. The size of the packet will be determined in advance, one gesture should take a certain number of seconds, depending on the speed of the user's gestures. Then the scenario of the model's work will be similar to the script of the sign language dactyl word recognition model. To ensure the work of the complex according to the scenarios, the following requirements were put to it.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 11:37:21 UTC from IEEE Xplore. Restrictions apply.

Mandatory requirements:

- The complex should work in real time, and also quickly and continuously process the data: take gestures, recognize them, send the finished message to the recipient, send the response to the user [5-6].
- All data must be stored on a cloud platform, so as not to occupy the local memory of the phone and do not slow down its operation [7].
- The complex should be able to adapt to the individual user and his gestures, and the user in turn should be able to train the program.
- It is possible to provide hot keys for calling special services and mandatory transfer of geolocation data to them.
- User must be able to record user contacts or synchronize them with a local device.

Additional requirements:

- The complex should have a convenient and pleasant interface, so that the user can easily understand its functionality, so that working with the application (the software package) was as comfortable as possible. Possibly changing the background color, its brightness and saturation. It can also provide a version of the interface for children with more vivid colors and patterns on the background. Perhaps, provide an interface for the visually impaired with bigger text and buttons.
- Possibility of text communication, that is, built-in instant messenger.

III. Creating a Training Set

The results of dactyl static sign recognition of people with hearing impairments were presented in the article. For learning the three previously selected intelligent algorithms of image recognition, namely knn, SVM and logistic regression, a training set of each sign images separately is needed. These images must meet the following requirements:

1. The sign should be correctly represented, according to how people with hearing impairment depict it.
2. Nothing should be presented on the image except the hand that creates a sign.
3. The background should be as light as possible (preferably white) and homogeneous, as the program assumes the transformation of the background into black for better hand finding.
4. The quantity of images for each sign should be enough for good learning - more than 200 pieces.
5. Each sign should be shown from different sides and with differences in the light fall, angle of inclination, etc.
6. Each sign should be photographed with the participation of different people.

The training set, shown in Fig. 6 and Fig. 7, was created as follows: 5-6 videos for each static sign were shot by different people for 10 seconds each. These videos were transmitted to the segmentation program [8], which divided them into frames to get 200 frames from each video, that is, at a rate of about 20 frames per second. For ease of use, all letters are called Latin letters. The figure below in brackets is the transcription of the Latin alphabet in Russian.

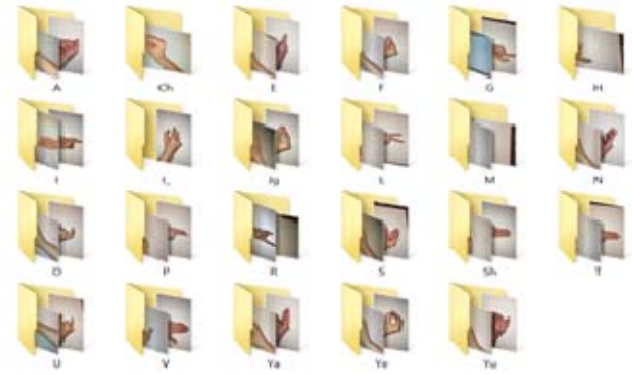


Fig. 6. The training set



Fig. 7. The Step of forming the training set for the letter "A"

Fig. 8 shows the structure of the project directory "Surdotelephone". The "code" folder contains the following scripts:

- (1) Putting the labels on the images by the name of the folders in order to make the model remember which images belong to which class of letters. The same pocedure is made for images from the training folder and the testing folder. For the testing folder, labels are used as a test of the correctness of the predicted value of the letter class for the image. It is compared with the label, so the program "understands" whether it was wrong;
- (2) Running the trained model (svm, k-NN or logistic) to recognize images from the testing folder (5);
- (3) The vector model (svm, k-NN or logistic), which vectors were created during the images transformation, memorizes the best weights and saves these values in the "output" folder (7) separately for each model;
- (4) Bringing the image to a size of 640 by 480 pixels, highlights the outlines of the hand, converts the

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 11:37:21 UTC from IEEE Xplore. Restrictions apply.

background of the image to black and translates the image into a pixel intensity vector;

In the "train" folder (6) there is a complete training set for models, and in the folder "test" (5) there is those images that were not included in the training folder and are needed for testing the models.

The "generated" folder contains all the generated program information, namely: the datas of the trained models (the "output" folder), the csv file that contains the pixel intensity vectors of all the images, and files that contain the information about the match of the images from the folders (5) and (6) their labels.

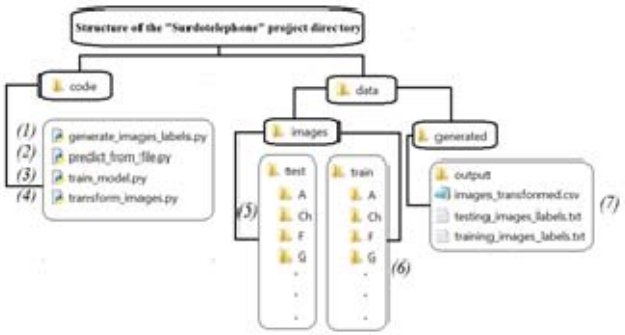


Fig. 8. The structure of the "Surdotelephone" project directories.

Programs for k-NN, SVM models and logistic regression partially use ready-made algorithms from machine learning libraries for Python [9].

IV. LEARNING THE ALGORITHMS AND WORK RESULTS

Below there is the description of actions sequence when the learning algorithm is working:

1. The Anaconda development environment is started, the environment previously created in it for the project is selected. The terminal, in which all the work will continue in the future, is opened. You need to go to the project folder, to the directory with the script code using the commands shown in Fig. 9.

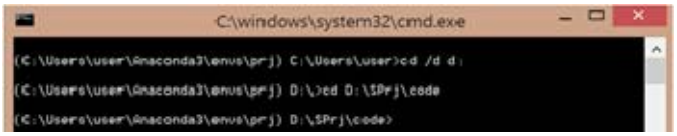


Fig. 9. The commands for transition to the necessary folder for starting the work with the recognition program.

2. Due to the Python command generate_images_labels.py train a link between the folder name and the photo for the train directory, which contains the training set of signs, is created. The process of the script working is shown below.



Fig. 10. The process of generate_images_labels.py script working.

This script forms the training_images_labels.txt file in the generated directory, which content partly reflected in Figure 11.

../data\images/train\A\001.jpg A

../data\images/train\A\002.jpg A

../data\images/train\A\003.jpg A

../data\images/train\A\004.jpg A

../data\images/train\A\005.jpg A

../data\images/train\A\006.jpg A

../data\images/train\A\007.jpg A

../data\images/train\A\008.jpg A

...

Fig. 11. The contents of the training_images_labels.txt file.

3. By running the transform_images.py script, all photos are converted to a single size, also the image is converted to a vector which values are in the range from 0 to 255, and this vector is saved to the image_transform.csv file in the generated folder. Fig. 12 shows how the program works for each image. Transforming about 7000 images takes about 10 minutes.

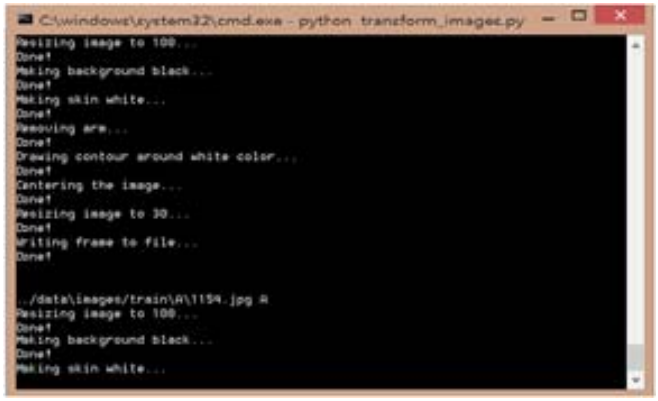


Fig. 12. The process of transform_images.py script working.

4. To train the model, you need to run the script train_model.py and through the spacebar the selected method knn, svm or logistic. The model training takes place on the csv file got in the previous step, on the generated vectors. After training, the model will be saved to the generated directory. The training time for the models is shown in Fig. 13.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 11:37:21 UTC from IEEE Xplore. Restrictions apply.

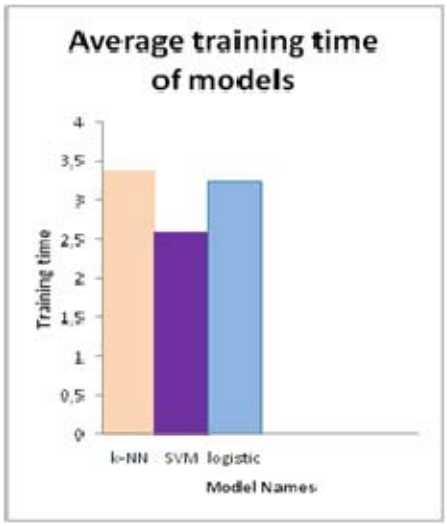


Fig. 13. The Comparative diagram of model training time.

The diagram shows that the model based on the method of supporting vectors has the best time of training.

For testing it is necessary to create directories in the test folder for each letter, similar to the training set, where the test images are placed. Next, it is necessary to run a script which puts the labels on all of the images in the test folder. It helps the program to determine whether the predicted value matches the correct label. Then, the predict_from_file.py model_name script is run. It checks all the images in the test folder. The testing process is shown in Figure 14.



Fig. 14. The process of the sign recognition programm testing.

The figure below shows how the program reacts to incorrect results.

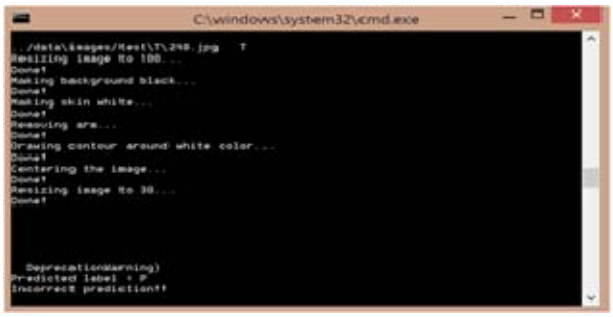


Fig. 15. The program console showing the error.

In this case, the label was "T", however the program (by the k-NN algorithm) recognized "P". These are Russian sign letters "T" and "P", which outlines are very similar.

The following results were gotten in the experiments:

Dactylems	Accuracy			Dactylems	Accuracy		
	logistic	SVM	knn		logistic	SVM	knn
А	92	94	85	С	90	93	86
В	93	94	83	Т	89	89	87
Г	92	92	87	У	94	92	94
Е	94	93	90	Ф	93	93	93
Ж	90	92	89	Х	92	94	92
И	89	92	86	Ч	92	92	84
Л	91	93	83	Ш	90	94	90
М	91	92	85	Ы	94	95	89
Н	92	95	88	Э	91	91	88
О	90	90	90	Ю	93	93	91
П	89	90	89	Я	93	94	80
Р	90	90	87				

Fig. 16. The accuracy of dactylic recognition models.

To evaluate the test results, the classification accuracy was calculated according to (1), where N is the number of correctly defined classes divided by the total number of test cases (size *(Xtest,1))*.

$$A = \frac{N}{size(Xtest,1)} * 100\% \quad (1)$$

As a result, the following datas were gotten: the method of reference vectors showed an accuracy of 92.5%, logistic regression - 91.5%, and the method of k nearest neighbors - 87.7%. Therefore, it is advisable to select the SVM method for further research and improvement.

Figure 17 shows the result of recognition of simple words by the reference vectors method.



Fig. 17. Recognizing the simple words "hello", "topic", "understood", "good".

The choice of these words is due to their frequent use in conversation with the teacher.

V. CONCLUSION

The article describes the development of a software package to support the distance education of people with hearing impairments. Scenarios of the levels of the recognition model for recognizing the complicating sign language structures, modeled in the BPwin program, are given.

The developed program complex has technical requirements.

The process of creating a training set of signs is described, the work of the recognition program is demonstrated. The results of experiments are analyzed, comparison of three models, their accuracy and speed of training is given.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 11:37:21 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] Zaitseva G.L. Sign Speech. Dactylology: Manual for student of supreme training institution. Moscow, VLADOS Publ, 2000. 192 p. (in Russian)

[2] Kleshnin N.G., Voronova L.I. NEURAL NETWORKS APPLICATION IN THE EMOTIONS RECOGNITION SUBSYSTEM FOR THE PROJECT "SURDOTELEPHONE" Telecommunications and Information Technologies, Moscow, 2018. T. 5. № 1. pp. 126-130

[3] Voronov V.I., Genchel K.V., Artemov M.D. and Bezumnov D.N., ""Surdotelephone" project with convolutional neural network," 2018 Systems of Signals Generating and Processing in the Field of on Board Communications, Moscow, 2018, pp. 1-6. doi: 10.1109/SOSG.2018.8350581

[4] Goncharenko A. A., Voronova L. I., Voronov V. I., Ezhov A. A. and Goryachev D. V., "Automated support system designing for people with limited communication," 2018 Systems of Signals Generating and Processing in the Field of on Board Communications, Moscow, 2018, pp. 1-7. doi: 10.1109/SOSG.2018.8350585

[5] Voronova L.I., Trunov A.S., Voronov V.I. THE DISTRIBUTED CALCULATORS MODEL FOR MOLECULAR-DYNAMIC SIMULATION OF STRONG INTERACTION SYSTEMS. International journal of experimental education, 2013, № 12. pp. 82-88

[6] Trunov A. S., Voronova L. I., Voronov V. I., Sukhachev D. I. and Strelnikov V. G., "Legacy applications model integration to support scientific experiment, " 2018 Systems of Signals Generating and Processing in the Field of on Board Communications, Moscow, 2018, pp. 1-7. doi: 10.1109/SOSG.2018.8350641

[7] Trunov A.S., Voronova L.I., Voronov V.I., Ayrapetov D.P. Container cluster model development for legacy applications integration in scientific software system (unpublished).

[8] FFmpeg – program for sequencing of frames by Microsoft. Available at: https://ffmpeg.org/ download.html#build-windows (accessed 11 July 2018)

[9] Conor Dewey. The Hitchhiker's Guide to Machine Learning in Python Medium member since. Data Scientist & Writer. Apr, 2018

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 11:37:21 UTC from IEEE Xplore. Restrictions apply.